

GIMLI HARBOUR, MB, CanadaWMO: **715770**Lat: **50.631N**Lon: **96.982W**Elev: **712**StdP: **14.32**Time Zone: **-6.00 (NAC)**Period: **99-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.4	-17.9	-28.6	1.1	-21.9	-24.5	1.4	-17.4	25.7	12.6	23.6	12.1	6.0	310	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	82.4	68.9	79.5	67.7	76.9	66.9	73.2	78.0	71.0	76.3	69.1	74.3	10.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	120.1	75.9	69.2	110.1	74.2	67.2	102.7	72.4	37.3	77.9	35.4	76.5	33.7	74.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.1	22.0	19.7	-26.6	89.2	4.8	3.0	-30.1	91.4	-32.9	93.1	-35.6	94.8	-39.0	97.0		
(5)			-26.9	77.5	4.7	2.4	-30.3	79.2	-33.0	80.6	-35.7	81.9	-39.1	83.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	4.0	6.6	20.8	35.4	49.4	61.9	68.2	66.0	56.8	42.9	27.2	10.0	
	DBStd	24.64	13.92	12.54	12.83	9.49	8.65	6.48	5.39	5.76	7.37	7.70	10.91	12.94	
	HDD50	6301	1425	1214	906	442	118	2	0	0	22	247	686	1239	
	HDD65	10275	1890	1634	1371	888	487	132	31	58	261	684	1135	1704	
	CDD50	1777	0	0	0	4	98	361	563	497	226	28	0	0	
	CDD65	277	0	0	0	0	3	40	130	90	14	0	0	0	
	CDH74	1327	0	0	0	0	37	179	643	412	56	0	0	0	
	CDH80	224	0	0	0	0	7	29	106	73	9	0	0	0	
Wind	WSAvg	9.5	8.7	8.3	8.8	9.4	10.5	9.5	8.8	9.4	10.3	10.6	10.4	8.9	
Precipitation	PrecAvg	21.5	0.9	0.6	0.9	1.1	2.5	3.6	3.1	2.9	2.3	1.7	1.1	0.9	
	PrecMax	29.6	2.3	1.1	3.0	2.4	4.7	7.7	6.7	5.8	4.6	3.6	3.3	2.2	
	PrecMin	14.1	0.3	0.0	0.1	0.4	0.7	1.5	0.7	0.6	0.7	0.4	0.2	0.2	
	PrecStd	3.8	0.5	0.3	0.6	0.5	1.1	1.7	1.2	1.2	1.0	0.9	0.6	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.7	40.2	51.7	65.2	79.1	83.7	87.5	86.3	80.1	67.4	54.0	37.5	
		MCWB	35.9	36.8	45.3	50.3	57.3	67.5	71.3	70.6	64.0	56.0	46.4	34.3	
	2%	DB	33.1	35.9	44.4	56.4	70.6	79.0	82.8	81.6	74.3	61.5	47.6	33.3	
		MCWB	31.9	33.5	39.8	46.2	57.1	64.5	69.8	68.9	64.1	54.0	42.9	31.7	
	5%	DB	28.6	31.2	40.4	50.9	65.1	75.1	79.9	78.2	70.8	58.0	43.9	31.0	
		MCWB	27.8	29.5	37.4	44.6	55.5	64.2	68.9	68.1	64.2	51.9	41.0	30.0	
	10%	DB	24.4	25.6	37.3	46.8	61.1	71.8	77.5	75.6	67.8	54.5	40.7	28.1	
		MCWB	23.5	24.1	35.1	42.3	54.4	64.0	68.1	67.0	62.1	49.3	38.2	27.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	37.5	45.6	51.7	63.5	73.5	77.4	75.3	70.6	58.5	47.5	34.8
			MCDB	38.6	39.7	49.4	60.2	67.2	79.5	81.8	79.3	72.9	63.5	51.9	37.0
2%		WB	31.7	33.7	40.6	48.0	59.9	69.2	74.1	72.5	68.1	55.6	43.7	32.1	
		MCDB	33.1	35.7	43.8	54.6	65.7	74.1	78.8	77.6	70.6	60.0	46.7	33.3	
5%		WB	27.5	29.1	37.3	45.3	57.3	67.0	71.8	70.6	65.5	52.7	41.0	30.0	
		MCDB	28.3	30.9	40.1	50.7	62.5	71.4	77.2	76.2	69.3	57.2	43.8	31.0	
10%		WB	23.3	23.9	35.1	42.1	54.8	65.1	69.8	68.7	62.6	49.4	38.3	27.1	
		MCDB	24.2	25.2	37.3	46.8	60.7	70.1	75.1	74.2	66.9	54.0	40.6	28.0	
Mean Daily Temperature Range	5% DB	MDBR	15.3	17.0	16.8	14.8	14.2	13.3	14.8	14.9	13.7	11.4	11.0	12.9	
		MCDBR	17.0	17.8	16.5	20.5	21.7	18.6	18.0	18.8	18.6	17.0	14.1	14.2	
		MCWBR	16.0	15.7	12.6	12.5	11.1	9.9	9.9	10.4	10.9	10.5	10.3	13.3	
	5% WB	MCDBR	16.7	17.7	15.8	20.6	17.2	15.0	15.3	15.9	14.6	14.8	13.1	14.4	
		MCWBR	15.4	15.7	12.2	13.0	10.2	9.8	10.5	10.5	10.3	9.8	9.9	13.3	
Clear-Sky Solar Irradiance	taub		0.264	0.296	0.305	0.354	0.368	0.384	0.394	0.377	0.346	0.301	0.271	0.247	
	taud		2.219	2.137	2.283	2.335	2.373	2.355	2.358	2.406	2.479	2.559	2.420	2.285	
	Ebn at Noon		242	261	285	282	282	277	272	271	269	266	244	234	
	Edh at Noon		22	32	34	36	36	37	37	33	28	21	18	18	
All-Sky Solar Radiation	RadAvg		387	759	1165	1597	1729	1835	1913	1590	1125	653	350	281	
	RadStd		39	55	87	108	147	144	88	112	101	84	50	28	